Digital Object Identifier

NYC Park Finder Individual Report : Enhancing Urban Well-Being through Predictive Park Recommendations

ADI BHATTACHARYA
School of Computer Science, University College Dublin, Belfield, Dublin 4, D04 V1W8, Republic of Ireland
adi.bhattacharya@ucdconnect.ie

I. INTRODUCTION

A. OVERVIEW

Problem: New York City, with its constant activity and noise, can be overwhelming, contributing to high levels of stress, adverse health conditions, and poor mental wellbeing.[1]

Solution: NYC ParkFinder is a web application designed to help users explore parks in Manhattan, NYC, offering personalized recommendations based on user preferences and predictions for park busyness using machine learning models. The platform aims to inspire outdoor activities, improving both physical and mental health by providing accessible urban green spaces for relaxation, physical activity, and social interaction.

Purpose: The primary purpose of NYC ParkFinder is to enhance user decision-making and encourage outdoor experiences by providing valuable information about Manhattan's parks, outdoor spaces, and events.

B. SCOPE AND OBJECTIVES

- **User-Friendly Interface:** Develop an intuitive and aesthetically pleasing interface to provide easy access to information about parks and events.
- **Personalized Recommendations:** Utilize user inputs and preferences to suggest suitable parks and green spaces.
- **Busyness Predictions:** Implement machine learning models to predict park busyness, helping users choose less crowded times for visits.
- **Future Enhancements:**
  - Safety and Crime Detection: Integrate safety features, using advanced data preprocessing to assess crime data accurately.
  - Interactive Heatmaps: Offer detailed insights into park busyness and safety through real-time data integration.
  - Route Planning: Enhance the map feature to provide routes from the user's location to recommended parks, increasing usability.
  - Mobile App Development: Create a mobile app for on-the-go access.
  - User Guidance: Implement instruction pop-ups and tool-tips to guide new users and notify them of new features.

II. MY UNDERSTANDING AND AIMS FOR THIS PROJECT

The NYC ParkFinder project addresses the challenge of finding quiet and suitable green spaces in the bustling urban environment of New York City. This difficulty is common in many cities, where residents and visitors seek not only tranquil areas but also specific amenities and activities to enhance their park visits. The application aims to provide users with the ability to select parks based on busyness levels, available activities, and amenities, thereby meeting their diverse requirements. By offering real-time insights and personalized recommendations, NYC ParkFinder helps users plan their visits to parks and events efficiently, catering to those looking for both serene and bustling environments.

A. ENHANCE DATA ANALYTICS AND MACHINE LEARNING UNDERSTANDING

- Develop a deep understanding of data analytics and machine learning models, particularly their performance and application in predicting park busyness.
- Integrate machine learning models with the backend to predict park busyness at specific times and dates.

B. SKILLS DEVELOPMENT IN DATA INTEGRATION AND PROCESSING

- Gain expertise in integrating various datasets to provide comprehensive and accurate information to users.
- Learn advanced data preprocessing and noise reduction techniques to handle complex datasets effectively.

C. IMPLEMENTATION OF REAL-TIME PREDICTIONS

- Implement real-time predictions and integrate them into heatmaps, providing users with detailed insights into park activity levels and safety.

D. RESEARCH AND DATA UTILIZATION

- Improve research skills to leverage current academic studies for informed decision-making regarding data visualization and machine learning models.

E. PROJECT MANAGEMENT EXPERIENCE

- Gain experience in managing a web application project from conception to deployment, including planning, development, testing, and maintenance phases.

F. COLLABORATION AND TEAMWORK

- Adopt best practices in collaborative development environments, working with a team to achieve project goals, which is essential for a professional work setting.

G. PROMOTE OUTDOOR ACTIVITIES

- Encourage users to spend more time outdoors by providing comprehensive information about parks, activities, and events, fostering a healthier lifestyle.

By achieving these aims, NYC ParkFinder strives to create a valuable tool that enhances the user experience, promotes outdoor activities, and contributes to the overall well-being of New York City residents and visitors.

III. TEAM STRUCTURE

The NYC ParkFinder project was executed by a multidisciplinary team organized to maximize efficiency and collaboration.

The team communicated primarily through Confluence for managing Scrum tasks and Slack for real-time communication.

A. COORDINATION LEAD

- Oversaw project timelines, deliverables, and stakeholder communication.
- Facilitated weekly meetings and ensured all team members were aligned with the project goals.
- Assisted in working on preprocessing and geocoding.

B. FRONT-END CODE LEAD

- Designed and implemented the user interface.
- Ensured the application was intuitive and visually appealing.

C. CUSTOMER LEAD

- Conducted rigorous testing to ensure the application met quality standards.
- Created prototypes and design elements to enhance user experience.
- Provided feedback to improve the aesthetics and information on the project video.

D. BACK-END CODE LEAD

- Developed server-side logic, integrated machine learning models, and managed database interactions to show busyness levels.

E. MAINTENANCE LEAD

- Minimized downtime and disruptions, ensuring that all systems and equipment were operating efficiently.
- Maintained the GitHub repository, ensuring that all maintenance documentation, manuals, and procedures were up-to-date and easily accessible for the team.

F. DATA LEAD (ME)

- Managed data acquisition, analysis, and machine learning tasks.
- Addressed challenges related to data quality, noise reduction, and model performance.
- Worked closely with the backend lead to implement busyness scores that connect to the backend to showcase park busyness, requiring new learning as I had limited experience with backend integration.
- Sourced, scripted, edited, and delivered the project video, ensuring it met all project objectives and quality standards.

IV. COMMUNICATION AND COLLABORATION

Throughout the project, I closely collaborated with the Coordination Lead, Ja Wei, on making critical decisions related to data, especially about geocoding and data preprocessing, as well as how to divide taxi zones according to the parks. Ja Wei's expertise and guidance were instrumental in refining our data strategy and ensuring the accuracy of our geospatial analysis.

Working with the Backend Lead, Yanwen, was also a crucial part of my role. Myself and Yanwen decided to sit down and make a plan for how we would integrate the components on the backend after the machine learning. We developed a comprehensive strategy to encapsulate all the work since the backend is in Java. Yanwen's collaboration skills were exceptional; she was patient, regularly solved our doubts, and ensured that our machine learning models were seamlessly integrated with the backend infrastructure.

In addition to the backend integration, we made sure to run these ideas by our Coordination Lead, Ja Wei, and Maintenance Lead, Shaline. Together, we drew up plans on how the machine learning could show the busyness levels under the parks. We also had to discuss how to do this differently for our secondary feature of a heatmap.

These discussions were pivotal in aligning our team's efforts and ensuring that the final implementation met all project requirements.

David, the Customer Lead, was instrumental in helping me improve the project video. His feedback on the aesthetics and information presented in the videos significantly enhanced their quality and ensured they met the project's communication objectives.

Laura, the Frontend Lead, created an amazing website that covered all the functionalities outlined in our initial mockup and discussions. Her ability to provide a helping hand whenever I faced difficulties regarding the busyness levels on the front end was invaluable. Laura's design and implementation ensured that the user interface was intuitive and visually appealing, greatly contributing to the overall user experience.

A. DECISION MAKING

- Decisions were made collaboratively, with input from all relevant stakeholders.
- Critical decisions were documented in Confluence to maintain a clear project history.
- As Data Lead, I played a key role in making decisions related to data strategy, data selection, and model selection.
- Major decisions were unanimously made ,however, decisions regarding a particular domain were by their respective domain lead.

Reflecting on the NYC ParkFinder project, the collaborative effort of our multidisciplinary team was key to its success. Each member brought their expertise and dedication, resulting in a robust application that effectively serves its users.

V. ROLE AND CONTRIBUTIONS

The project began with an in-depth exploration of various datasets to define and predict park busyness effectively. The aim was to identify datasets that would provide comprehensive, timely, and relevant information to enhance user decision-making and improve the overall functionality of the NYC ParkFinder application. I used the following format to make my decisions :

- **Pros:** Useful for determining the occupancy and movement of people every minute.
- **Cons:** Location data is provided in TLC taxi zones (MULTIPOLYGON), which does not specify exact parts of Manhattan.
- **Final Decision:** Yes. However, we need to decide how to handle TLC taxi zones. Additional TLC datasets might be required to refine location data.

A. DATASET EVALUATION PROCESS

After the initial evaluation, various factors were considered to determine busyness in Manhattan, especially after dropping the idea of using crime rates. Here's a summary of the subsequent steps:

1) Refinement of Initial Ideas

Crime Rates:

- Initially considered identifying crime rates using available datasets.
- As the project scope evolved, the focus shifted away from crime scores to other factors influencing park busyness.

2) Exploration of Alternative Datasets

Weather Data:

- Investigated datasets providing weather information to understand its impact on park attendance and user preferences.

Turnstile Data:

- Examined subway turnstile data to gauge foot traffic patterns and predict peak times for park visits.

Event Data:

- Reviewed various datasets detailing events taking place in Manhattan to correlate with increased park busyness.

3) Challenges and Considerations

Data Accuracy and Timeliness:

- Ensured that selected datasets were accurate and regularly updated to provide reliable predictions.

Integration Feasibility:

- Evaluated the ease of integrating each dataset with the existing backend systems.
- Considered technical requirements and potential challenges in combining multiple datasets.

User Decision-Making Enhancement:

- Focused on datasets that could enhance user decision-making by providing actionable insights on park busyness and safety.

B. DATA PREPROCESSING AND INTEGRATION

- Began by utilizing various preprocessed datasets provided by Ja Wei, merging them with additional datasets such as weather data and park busyness metrics.
- Initial steps included converting categorical columns to the category data type and encoding the necessary columns to prepare the data for machine learning processes.
- This setup was essential for performing Principal Component Analysis (PCA) and implementing machine learning models.

C. DATA VISUALIZATION AND PCA ANALYSIS

Scatter Plots for Relationship Visualization:

- Utilized scatter plots to visualize the relationship between weather variables and park busyness. These visualizations helped identify patterns and correlations that could influence park attendance.

PCA Analysis:

- Conducted PCA with two components to identify the initial explained variance ratio.
- Explored additional principal components to capture a higher variance, revealing that approximately eight components were necessary to capture around 90% of the variance.
- This comprehensive approach ensured a thorough understanding of the data's dimensionality and variance distribution.

Evaluation of Principal Components:

- Checked the cumulative explained variance to determine how much variance was captured by the principal components.
- Ranked features based on their contributions to the PCA, providing insights into the most influential variables affecting park busyness.

By following these steps, the data was well-prepared for machine learning applications, allowing for accurate predictions and valuable insights into park busyness in Manhattan.

VI. GEOSPATIAL ANALYSIS FOR MODEL REFINEMENT

After determining the appropriate feature rankings, I collaborated with Ja Wei to understand how to set ranges around parks and perform geospatial analysis. This was a challenging task for me, as I had never worked on geospatial analysis before and initially found it difficult to grasp. We spent over two weeks trying to map park busyness using taxi zones, which was a significant challenge. This required us to carefully assess and determine the appropriate taxi zones for each park.

A critical error we encountered was the lack of a dataset that divided Central Park into its various small sections. This oversight potentially skewed the busyness scores, as Central Park's vast area was not adequately segmented.

A. GEOSPATIAL ANALYSIS STEPS

1) Initial Approach

We started by using the proportion of parks covered by taxi zones. However, after completing the training and predictions with the chosen model, we realized that this method was not optimal.

2) Revised Method

After recognizing the limitations of our initial approach, we switched to using percentiles of the parks, which improved the model's performance. We normalized the busyness scores, leading to more accurate and consistent predictions.

Ja Wei played an instrumental role in refining this process and ensuring that the adjustments were implemented effectively and on time. However, due to the time spent resolving these issues, our data analysis progress lagged behind other teams.

This iterative process of addressing geospatial analysis challenges and refining our approach was crucial in enhancing the overall functionality of the NYC ParkFinder application.

VII. MACHINE LEARNING

A. LINEAR REGRESSION FOR TIME SERIES FORECASTING

I began by constructing a linear regression model through a comprehensive modeling pipeline. This allowed me to make predictions that facilitated an initial evaluation of the linear regression model. I calculated several key performance metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared. The results indicated that most predicted values clustered near zero, regardless of the actual values, suggesting that the model failed to capture the variance in the data.

To enhance the evaluation of my predictive model, I implemented a custom loss function. The objective was to develop a more comprehensive metric that incorporates both MSE and MAE, while also introducing a penalty for larger errors. The custom loss function is defined as a weighted sum of MSE, MAE, and the large error penalty, with the formula:

$$\text{Custom\_loss} = 0.5 \times \text{MSE} + 0.25 \times \text{MAE} + 0.25 \times \text{large\_error\_penalty} \tag{1}$$

- **0.5 for MSE:** Ensures that large errors have a significant impact on the loss value, encouraging the model to minimize them.
- **0.25 for MAE:** Balances the influence of MSE by considering all errors equally, promoting overall accuracy.
- **0.25 for Large Error Penalty:** Specifically targets extremely large errors, penalizing the model for significant deviations.

The calculated loss value provided a more nuanced view of the model's performance, highlighting areas of strength and weakness, particularly in terms of large errors. This approach underscored the importance of considering multiple facets of error in model evaluation and inspired me to think more critically about tailoring evaluation metrics to specific project needs.

Given that the linear regression model, as currently specified, proved inadequate for predicting park busyness, I decided to explore Long Short-Term Memory (LSTM) networks as a potential solution.

B. EXPLORING LONG SHORT-TERM MEMORY (LSTM) NETWORKS FOR IMPROVED PREDICTIVE MODELING

Motivation: My primary motivation for choosing LSTM networks was their proficiency in capturing long-term dependencies in time series data through specialized memory cells and gating mechanisms.

First Attempt: Initial Model Training

Training Process:

- **Loss Values:** Initially, the loss values fluctuated and occasionally increased, indicating potential challenges in finding the optimal fit.
- **Early Stopping:** The early stopping callback activated after 10 epochs, suggesting that the validation loss had ceased improving, thereby preventing overfitting.

Results:

Test Loss: The model evaluation on the test set recorded a test loss of 5.1504, indicating an average prediction error of this magnitude.

Second Attempt: Model and Learning Rate Adjustments Dissatisfied with the initial results, I modified the model structure and learning rate

Training Process:

Stable or Reducing Loss: The training and validation losses showed slight reductions but remained nearly constant, indicating limited improvement with this configuration.

Third Attempt: Enhancing Generalization To further enhance the model's generalization ability, I introduced additional dropout and regularization techniques

Training Process:

- **Increased Dropout Rates:** Enhanced dropout rates to mitigate overfitting and improve generalization.
- **Regularization Techniques:** Implemented L2 regularization in LSTM layers to penalize overly complex models.
- **Patience Parameters:** Adjusted early stopping and learning rate reduction parameters to allow more learning time.

Results: These adjustments aimed to create a more robust model capable of better generalization to unseen data. This iterative process reinforced the lesson that machine learning involves constant tweaking, testing, and refinement to achieve optimal results.

Reflection: Throughout these experiments, I learned several foundational lessons:

1) The importance of monitoring loss values and using early stopping to prevent overfitting.
2) The significant impact of adjusting model architecture and learning rates.
3) The value of enhancing generalization through dropout and regularization techniques.

These experiences underscored that developing a successful machine learning model is an iterative process. Each experiment builds on the previous ones, offering new insights and guiding further improvements.

C. SUPPORT VECTOR MACHINE (SVM) MODELS FOR TIME SERIES FORECASTING

Motivation: While Long Short-Term Memory (LSTM) networks excel at capturing long-term dependencies in time series data when sufficient training data is available, they did not yield satisfactory results in this context. Therefore, we opted to use Support Vector Machines (SVM) due to their robustness against feature scaling and lower susceptibility to overfitting compared to other methods like neural networks.[2][3]

First Attempt: Exploring Different Kernels To start, I experimented with various kernels for the SVM model. Kernels are fundamental to how SVM models transform the data to find the optimal decision boundary.

Training Process:

- **Kernel Impact:** Different kernels (like linear, polynomial, and radial basis function) impacted the model's performance differently. This highlighted the importance of selecting the appropriate kernel based on data characteristics.
- **Generalization Issues:** The models struggled to accurately predict higher values, indicating potential generalization issues. This suggested the need for more sophisticated approaches or improved preprocessing.

Second Attempt: Hyperparameter Tuning To enhance performance, I conducted hyperparameter tuning for the best-performing kernel. Here's the approach I took:

Learning Outcomes:

- **Parameter Sensitivity:** SVM models are highly sensitive to hyperparameters like C, gamma, and epsilon. Finding the right combination is crucial for model performance.
- **Initial Disappointment:** Despite extensive tuning, the results did not significantly improve. This highlighted that even well-planned tuning might not always yield expected improvements.

Third Attempt: Refining the Approach Realizing the need for a more refined approach, I focused on enhancing data preprocessing and simplifying the hyperparameter search space. Here's the refined strategy:

Learning Outcomes:

- **Importance of Preprocessing:** This attempt reinforced the importance of thorough data preprocessing, including feature encoding and scaling, to improve model performance.
- **Focused Hyperparameter Tuning:** By narrowing the range of hyperparameters, I found a better configuration more efficiently. This demonstrated the value of a focused and informed search.
- **Improved Results:** The refined approach yielded better results, underscoring that iterative refinement and methodical adjustments lead to enhanced model performance.

Reflection: Through these experiments, I learned foundational lessons about machine learning and model development. Each step emphasized the importance of understanding the underlying mechanisms of models, the impact of preprocessing, and the iterative nature of refining a model. This project taught me that successful model development requires both rigorous testing and creative problem-solving.

VIII. LESSONS LEARNED FROM THE NYC PARKFINDER PROJECT

A. DATA ANALYTICS AND MACHINE LEARNING

- **Deepened Understanding:** This project significantly enhanced my understanding of data analytics and machine learning models, particularly their performance and application in predicting park busyness.
- **Advanced Preprocessing:** I learned advanced data preprocessing and noise reduction techniques, which were essential for handling complex datasets effectively.

B. GEOSPATIAL ANALYSIS

- **Geospatial Mapping:** Collaborated with Ja Wei to set ranges around parks and perform geospatial analysis, mapping park busyness using taxi zones. This experience was challenging but crucial for understanding spatial data.
- **Refinement Process:** The process involved several iterations and method refinements, leading to the realization that percentiles of park areas provided a more accurate model than the initial proportional approach.

C. PROFESSIONAL SKILLS

- **Collaboration and Teamwork:**
  - **Cross-Functional Collaboration:** Worked closely with team members across different functions, such as frontend and backend development, ensuring seamless integration of machine learning models with the application's backend.
  - **Effective Communication:** Utilized Confluence for managing Scrum tasks and Slack for real-time communication, facilitating transparent and structured collaboration.
  - **Problem-Solving:** Learned to address challenges related to data quality, integration, and model performance through iterative refinement and continuous improvement.
- **Research and Application:**
  - **Academic Studies:** Improved research skills by leveraging current academic studies to make informed decisions regarding data visualization and machine learning models.
  - **Practical Application:** Applied theoretical knowledge to practical scenarios, enhancing the application's functionality and user experience.
- **User-Centric Design:**
  - **Intuitive Interface:** Focused on developing a user-friendly and aesthetically pleasing interface, ensuring easy access to information about parks and events.
  - **Personalized Recommendations:** Implemented features that provide personalized park recommendations and busyness predictions, catering to diverse user preferences and needs.

D. FUTURE APPLICATIONS

- **Continued Learning:**
  - **Advanced Machine Learning:** Plan to further explore advanced machine learning techniques and their applications in real-world scenarios, aiming to avoid overfitting and find efficient ways to predict values.
  - **Geospatial Analysis:** Intend to delve deeper into geospatial analysis and understand its applications, as I found this challenging and currently have only a basic understanding of it.
- **Professional Development:**
  - **Project Management:** Aim to apply the project management skills learned to future projects, ensuring efficient planning, execution, and continuous improvement.
  - **Collaboration:** Will continue to adopt best practices in collaborative development environments, working effectively within cross-functional teams to achieve project goals.
- **User Experience Enhancement:**
  - **Feedback Integration:** Plan to integrate user feedback more systematically to refine features and improve the overall user experience.
  - **Mobile Application Development:** Explore the development of mobile applications to increase accessibility and provide on-the-go functionality.

The NYC ParkFinder project provided a comprehensive learning experience, combining technical skill development with professional growth. These lessons will be invaluable in my future endeavors, both in technical roles and collaborative projects, ultimately enhancing my ability to contribute effectively to any team or project.

IX. REFERENCES

1) NYC Department of Health and Mental Hygiene. (n.d.). Noise [Online]. Available: https://www.nyc.gov/site/doh/health/health-topics/noise.page
2) Anonymous. "Which performs better in time series forecasting: LSTM or SVR," Data Science Stack Exchange, Dec. 19, 2018. [Online]. Available: https://datascience.stackexchange.com/questions/54657/which-performs-better-in-time-series-for
3) M. K. Okasha. (2014, Jan.). "Using Support Vector Machines in Financial Time Series Forecasting." International Journal of Statistics and Applications [Online]. 4(1), pp. 28-39. Available: http://article.sapub.org/10.5923.j.statistics.20140401.03.html